Monte Carlo Modeling of Ostwald Ripening

EN510.666 Materials Modeling

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Abstract:

Ostwald ripening is a well-known thermodynamic-driven process that impacts the evolution of dispersed phases in a wide-range of materials systems. For this project, we developed a simplified kinetic Monte Carlo (KMC) model to investigate Ostwald ripening under detachment-limited conditions. The model represents dispersed phase particles as aggregates containing a discrete number of "atoms" that can "dissolve" and go to another particle. Detached atoms can either be assigned randomly or by weighted assignment that is proportional to the surface area. A variety of simulations were performed with different numbers of particles, initial size distributions, critical radii, numbers of iterations, and atom assignment methods. The results show that the simple model reproduces key qualitative aspects of Ostwald ripening such as the preferential dissolution of smaller particles, reduction in total number of particles, and change in particle size distribution. Moreover, the mean particle radius follows the geneal theory established by Lifshitz-Slyozov-Wagner (LSW), but with slower rates due to the simplifying assumptions of the model. While the algorithm has issues scaling to a large number of particles, the code runs within a reasonable time frame for most simulations. Future work should include adding more physics to the simplified model such as diffusion, and particle interactions.

1 Introduction

1.1 Ostwald Ripening

Ostwald ripening is a thermodynamically-driven process that causes small crystals or particles to dissolve and migrate towards larger crystals or particles over time [1]. The process was first described by Wilhelm Ostwald in 1896 who noticed that the largest crystals precipitating from a solution tended to grow while the smallest ones shrank [2]. This effect was formalized with the Ostwald–Freundlich equation which relates the effective vapor pressure to the surface curvature (particle radius) as shown in Equation 1. Here, γ is the surface tension, V_m is the molar volume, r is the particle radius, R is the universal gas constant and T is temperature. By assuming an ideal solution, the vapor pressure can be related to the to concentration in a solid solution. Thus, Equation 1 can transform into Equation 2 where $C(r)$ is the solubility for a particle with radius r and C_∞ is the bulk concentration.

$$\ln\left(\frac{p}{p_{sat}}\right) = \frac{2\gamma V_m}{rRT} \tag{1}$$

$$C(r) = C_\infty \exp\left(\frac{2\gamma V_m}{rRT}\right) \tag{2}$$

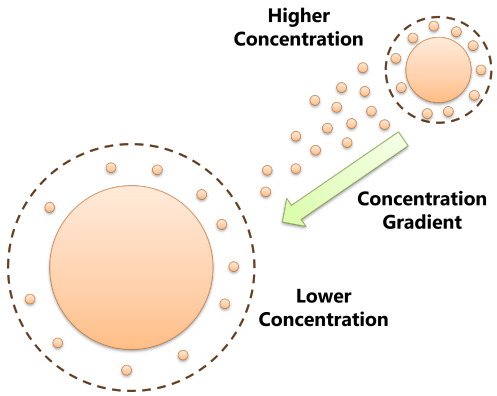


Figure 1: Schematic showing a simplified version of Ostwald ripening between two particles. Smaller particles will have high chemical potential, and consequently a higher equilibrium concentration which produces a concentration gradient between the large and small particle.

From Equation 2, we can see that smaller particles will have an higher effective solubility in solution compared to larger ones. The difference in solubility between small and large particles creates effective concentration gradients which leads to solute diffusing away from smaller particles and towards larger ones. This process is schematically shown in Figure 1. Ostwald ripening can also be thought of lowering the total energy of the system by reducing the total amount of interfacial area (and consequently energy). Ostwald ripening is applicable to many liquid-vapor and solid-liquid systems; for example in age hardened alloys like AlMgSi [3], emulsions [4], organic biomacromolecules [5], batteries [6], and many nano-structured materials [7]. Having a better fundamental understanding of Ostwald ripening could help improve areas where ripening yields negative effects such as nanoparticle catalysts failing at high temperatures due to sintering [8].

Many researchers have previously modeled Ostwald ripening using computational methods. Predominantly, numerical methods such as phase-field modeling have been employed [9–13]. Phase-field modeling describes a system using a continuous order parameter that defines the local phase at each discretized point in space, with diffuse ("fuzzy") interfaces between phases rather than sharp boundaries [14]. The evolution of the system is governed by a free-energy functional, which describes the total system energy and is solved iteratively through a set of partial differential equations at each time step [14]. Thus, phase-field modeling is useful for modeling Ostwald ripening since the process involves the evolution of the phase boundaries between two phases.

Here, we model Ostwald ripening using a highly-simplified Monte Carlo approach. Before discussing our implementation, we briefly cover Monte Carlo methods and its applicability to modeling various material science related problems.

1.2 Monte Carlo

Monte Carlo refers to a set of computational algorithms that utilize repeated random sampling to produce numerical results for problems with inherent uncertainty or stochasticity. It is used when events in the system occur according to some probability distribution or at a certain rate. The algorithm uses randomness and many different trials to estimate a range of outcomes when exact math cannot be used.

Pedagogical toy problems include approximating pi, estimating integrals, or simulating random walks. The first two incorporate Monte Carlo by the random generation of points across a region, and the fraction of points that fall within a circle or integral curve is used to estimate pi or area under a curve. The estimation accuracy increases with the number of points generated. Random walks are simulated by randomly choosing a step direction at each time point, and this is used to calculate metrics such as displacement from the origin as a function of time.

Classical Monte Carlo tends to simulate an equilibrium state, often running relaxation trials from an unstable initial state to a stable equilibrium, providing data about the final system configuration. On the other hand, Kinetic Monte Carlo (KMC) focuses on the time evolution of a system by randomly selecting events based on their associated rates. KMC is useful when the system is not ideal and cannot be characterized by exact equations. This makes analytical solutions inaccurate or difficult to calculate without complicated error propagation. KMC is also useful when the time-scales required to reach equilibrium are long and cannot be easily simulated using Molecular Dynamics.

Monte Carlo has been used extensively to model material systems such as copolymer glass-rubber interfaces [15] and grain growth in polycrystalline crystals [16].

Work by Baeurle investigated the material properties of

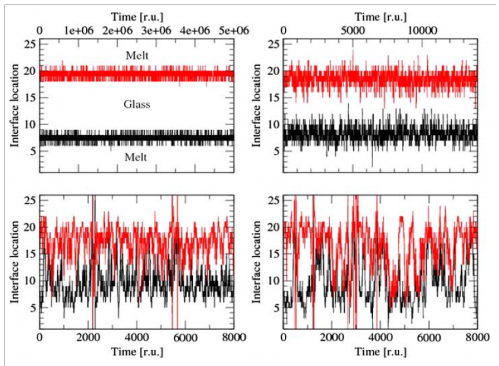


Fig. 9 Glass–rubber interface locations versus real time at various temperatures for a lamellar TPE material composed of SIS triblock copolymers, determined with the combined KMC–SCFT algorithm. In the calculations the following temperatures have been considered: $T = 305$ K (upper left), $T = 320$ K (upper right), $T = 350$ K (lower left) and $T = 380$ K (lower right)

Figure 2: Glass-melt interface locations as a function of time at different temperatures [15]

block-copolymer-based TPEs with alternating glass-rubber phases under extensional strain 2. They used KMC to melt or freeze units at the interface with certain probabilities that varied according to temperature, and found that higher temperatures caused higher interface fluctuations, with the glassy phase being completely molten at the highest temperature (T = 380 K).

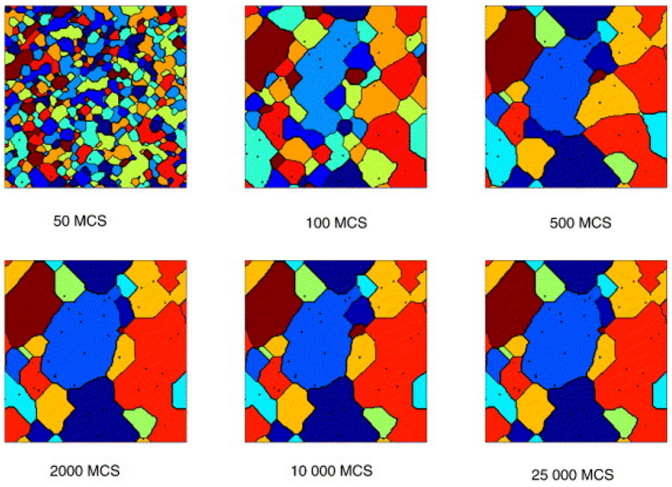


Figure 3: Grain growth in a polycrystalline materials after different numbers of Monte Carlo steps [16]

Work by Ming Huang et al. used KMC to model grain growth in a polycrystalline material. They initialized a discrete lattice of points, each associated with a certain grain direction. At each step, they chose a random site to change orientation with a probability dependent on how it would change the system's interfacial energy [16].

2 Approach

2.1 Assumptions

To simplify the Ostwald Ripening scenario for modeling, we make two main assumptions about the relative kinetics of the system and the behavior of detached atoms.

First, we consider only detachment-limited Ostwald Ripening. This assumes that atom detachment from particles is slow relative to atom diffusion throughout the system and atom attachment to new particles. As a result, the kinetics of particle growth/shrinkage are moderated solely by the rate of detachment, which is a function of particle size. This is how Monte Carlo is integrated into our solution, where every particle loses an atom with some probability at each timestep. Atoms below a critical radius r_c lose an atom with certain probability, while atoms with radius larger than r_c lose an atom with a probability that exponentially decays as r increases. This implements accurate rates while incorporating the randomness seen in physical systems.

Second, we assume that each released atom immediately attaches to another particle. In choosing which particle gains an atom, we investigate two scenarios: an unweighted probability distribution (where each particle is equally likely to receive an atom), and a weighted probability distribution (where particles with higher surface area are more likely to receive a particle).

2.2 Code Methodology

To model this system computationally, three separate classes were implemented: Ostwald.Atom() Ostwald.Particle(), and Ostwald.System(). Together, they form the basis of the projects implementation, together making up for all mathematical operations as well as all internal bookkeeping.

2.2.1 Ostwald.Atom()

The Atom class is what aggregates to form a particle. In real systems, the constituent material need not be atoms. The reason an entire class was created for Atom is for potential future applications where different instantiations of atoms may be utilized (e.g. different charges, different volumes, or different masses). The Atom class is one which possesses no internal non-getter/setter methods. It exists to mimic atomic properties, and possesses the following internal properties: Atom.mass, Atom.vol, and Atom.charge. The properties reference mass, volume, and charge respectively. For the scope of this project, Atom.vol is the only attribute utilized, but the other attributes are still potentially relevant to other fields of study as mentioned prior.

2.2.2 Ostwald.Particle()

The Particle class is what houses the aggregate material. Similarly to in the Atom class, the constituent material need not be atoms. In this implementation, Atom, and Particle are nomenclature which allow the forces and processes involved to be intuited cleanly. Particle contains more methods than Atom, as it is responsible for keeping track of the constituent Atom classes which it is comprised of. Upon instantiation, an object with the Particle class will be created with a position on the xyz plane (z is defaulted to 0), and an initial list of atoms (likewise defaulted to empty).

In addition to getter and setter methods, the Particle class possesses two methods which are important to the Sys-

tem class: Particle.addem(atom) and Particle.losem(index). These two methods add an object of the Atom class and remove one at a given index, respectively. For the purposes of data and system implementation, there also exist internal methods to calculate radius, surface area, and net charge. The former two methods rely on the assumption that the volume of every constituent Atom class summed is equal to the volume of the particle, and that the particle's shape is approximately spherical. This assumption is made so that the formula for a sphere can be used in tandem with a method to calculate total volume by summing up the volume of internal atoms.

2.2.3 Ostwald.System()

The System class is what possesses the majority of internal logic regarding this simulation. Similarly to how the Particle class possesses a list of objects of the Atom class, the System Class possesses a list of objects of the Particle class. Inside of Ostwald.System() are a large variety of ways to manipulate the constituent Particle instances. To begin, the System instantiation can be specified to possess dimensions, a critical radius, and a list of particles. Each parameter defaults to the None type, 1.1, and an empty list, respectively, if not specified. Existing inside this class are methods to record particle data, iterate a given number of time steps, create particles, perform sweeping operations on particles, and export data as a csv file. To begin from scratch, a standard System must be build with System.buildStdSystem() with parameters dictating the dimensions in which Particle objects can initially be found inside, as well as the number of total Particle objects to be instantiated.

With the System class's internal logic, it may also be expanded upon in the future if the user sought to create any further logic in the simulation. As mentioned in the description of the Atom class, there are multiple factors that could be accounted for regarding Particle composition and methodology. One such factor is the addition of motion in a creation of a three-body problem utilizing charge and Ostwald ripening to mimic solid solutions or droplets in space (such as two non-mutually soluble mixtures with constituent particles).

2.3 Monte Carlo Implementation

As mentioned before, this project primarily relies on Kinetic Monte Carlo simulation. Inside the System class there are two methods which employ this, System.iterate() and System.getWeightedIndex(). System.iterate() is the method used to iterate through every Particle inside the list and decide whether or not each will lose an Atom, and then again how the lost Atom objects will be distributed. In order to utilize Monte Carlo randomness, System.iterate utilizes a piecewise function given by the following:

$$P(r) = \begin{cases} 1: & 0 \leq r < 1 \\ e^{\left(1-\frac{r}{r_c}\right)}: & 1 \leq r < \infty \end{cases} \tag{3}$$

Where $P(r)$ represents the probability of a Particle with radius r and System with critical radius r_c to lose a constituent atom. This formula removes many of the constants mentioned prior, but from a modeling position we can observe, and will discuss later, that these assumptions are sufficient to provide a system which reflects what is experimentally observed.

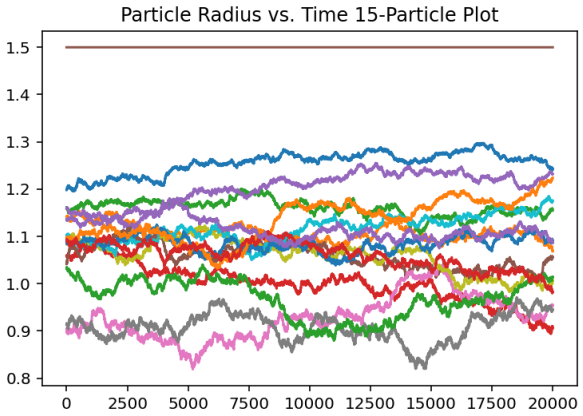


Figure 4: 15-particle plot with 20,000 iterations performed without index weighting at critical radius 1.5.

Here, the Monte Carlo method is called upon to randomly generate numbers between 0 and 1, and then compare them to the value output by Equation 3 (for larger simulations, only one Monte Carlo iteration is performed as the vast number of Particle objects accounts for the lost certainty and otherwise would prove computationally expensive). The reason the ratio is used as oppose to a difference is that a ratio allows for the system to be more scalable, allowing for a wider spread of particle sizes without needing further computations for critical radius difference.

2.3.1 Effect of Index Weighting

Within the System.iterate() method, there exists the option of choosing a Particle to receive an atom which has been lost by another Particle either by random, or by weighted a random probability. To illustrate the value of Monte Carlo Randomness, Figures 4 and 5 display two identically seeded systems. Figure 4 displays a system where each Particle that is chosen to receive an atom is done so randomly. To contrast, Figure 5 uses a Monte Carlo sweep to decide, based on weighted surface areas of each Particle, which one will receive the atom. The aforementioned data should display how important it is to utilize some form of weighting or other mathematically derived algorithm which makes physical sense.

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3 Results & Discussion

3.1 Individual Particle Growth Curves

Figure 6 shows individual particle growth trajectories for a 200-particle simulation over 40,000 time steps with weighted atom assignment. The plot reveals that particles that are initially smaller than the critical radius will usually become

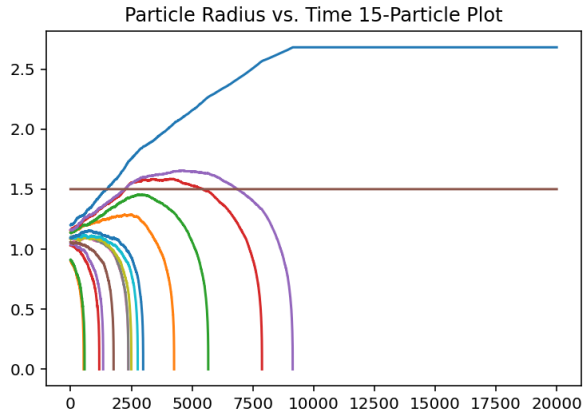


Figure 5: 15-particle plot with 20,000 iterations performed with index weighting at critical radius 1.5.

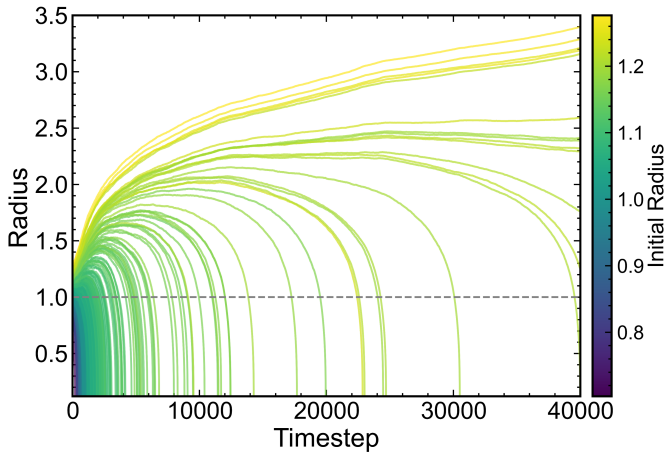


Figure 6: Individual particle growth trajectories for a 200-particle simulation over 40,000 time steps with a weighted atom assignment. Each line corresponds to a single particle growth trajectory. The color for each particle represents its initial radius and the dashed grey-line represents the critical radius for this specific simulation ($r_{crit} = 1.0$).

smaller, which makes sense given the fact that the code makes the loss probability = 1 for particles that are smaller than the critical radius. However, it is still possible that particles below the critical radius will grow under some simulation conditions. This is because although they will always lose and atom, they also have a probability of gaining atoms as seen in Figure 5. Considering the simulation seen in 5 with only 15-particles, we see that a few particles that are below the critical radius of 1.5 do in fact grow initially. This simulation implies that Ostwald ripening will occur even when particles are initially formed and unstable. In this simulation, particles will grow until they surpass the unchanging r_{crit} as mass flow between particles is conserved.

Looking back at the larger-scale simulation shown in Figure 6, we see that the majority of the small particles last less than 5,000 time steps. Particles that are slightly larger than the critical radius initially grow, but once they become the smallest particles, they tend to give up atoms to larger particles. Once particles reach their maximum size, they tend to shrink exponentially until totally dissolving.

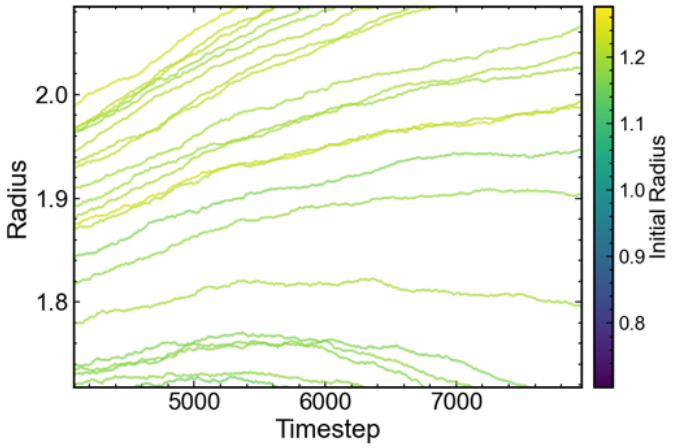


Figure 7: A zoomed in plot for the same 200-particle simulation over 40,000 time steps with a weighted atom assignment. Each line corresponds to a single particle growth trajectory. The color for each particle represents its initial radius.

By the end of the simulation, very few particles are remaining. This demonstrates how Ostwald ripening can lead to a drastic reduction in the total overall number of particles in the system. Figure 7 gives a zoomed in version of Figure 6 at a time point when the vast majority of smaller particles have dissolved. From the zoomed in plot, it is easier to see that for a single timestep, there is some stochasticity with whether particles grow or shrink. That is to say, although many larger particles tend to gain atoms over a long enough timescale, at short timescales, it is possible even for the largest particles to lose an atom. We also note that Figure 6 and Figure 7 demonstrate that some particles that are initially larger than others do not end up being larger by the end of the simulation.

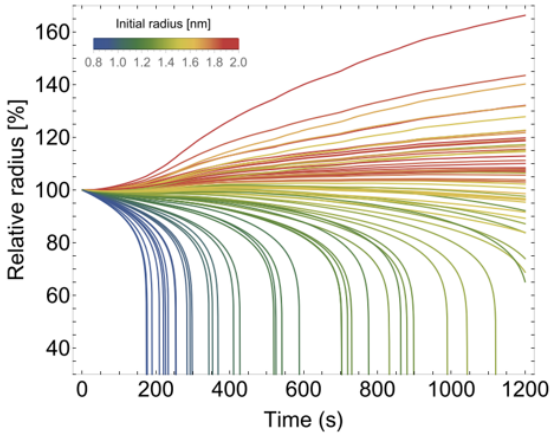


Figure 8: Plot taken from Bucci et al. [17] (see Figure 12(b) in original paper) that shows particle growth trajectories for many particles, color-coded based on their initial radius.

Our results can be compared to a similar plot from Bucci et al [17] who used mesoscale modeling to predict the coarsening of platinum-based nanoparticles in proton exchange membrane fuel cells. Figure 8 reprints Figure 12(b) from [17]. Although Bucci et al. used a different simulation method, and incorporated diffusion into their modeling, we can still see that our results generally agree with theirs.

3.2 Particle Size Distributions Over Time

Figure 9 plots particle size distributions for the same 200-particle simulation that was referenced in the previous section.

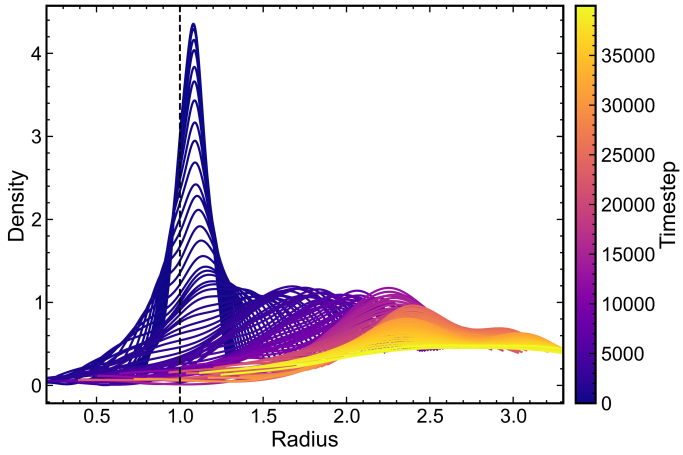


Figure 9: Kernel density estimate size distribution curves across various time steps in the 200-particle simulation over 40,000 time steps with a weighted atom assignment. The line color represents the distribution at the corresponding time step from the color bar, while the dashed black line represents the r_{crit}.

The time evolution of particle size distribution closely aligns with work from Bucci et al. Figure 11a. [17]. Despite their simulation using many more variables the similarity demonstrates that particle radius is a major factor for Ostwald coarsening. Both simulations begin as a Gaussian; overtime, the radius distribution increases as smaller particles merge into the larger ones. The main difference between our plot and theirs was the formation of a bimodal distribution with a peak at the smaller particles and a second, smaller peak for the large radii. This is very apparent when looking at the 25000th time step of Figure 9. We hypothesize the reason for this is due to the reduced overall absorption. In our simulation, a majority of particles dissipated very early on. Since each remaining particle could give at most one atom, the change in growth rate stalled, not creating the right-skewed distribution we would expect.

3.3 Mean Particle Radius Over Time

The mean particle radius, $\langle r \rangle$ models the average growth of the larger particles driven by curvature-induced chemical potential differences during Ostwald ripening. Lifshitz Slyozov Wagner (LSW) theory models this coarsening process showing that it is diffusion-limited and leads to a characteristic cube-root growth law. The radius distribution evolves linearly with time, such that

$$\langle r \rangle^3 - \langle r_0 \rangle^3 = Ct \tag{4}$$

where

$$C = \frac{8\gamma c_\infty v^2 D}{9 R_g T} \tag{5}$$

(The interfacial energy γ, solubility c_∞, molecular volume v, diffusivity D, and thermal energy).

As a result, $\langle r \rangle$ is a very good descriptor of coarsening kinetics and has been widely used to compare simulation and experimental results. This equation is especially helpful as it is applicable across different systems, even when absolute length and time scales differ.

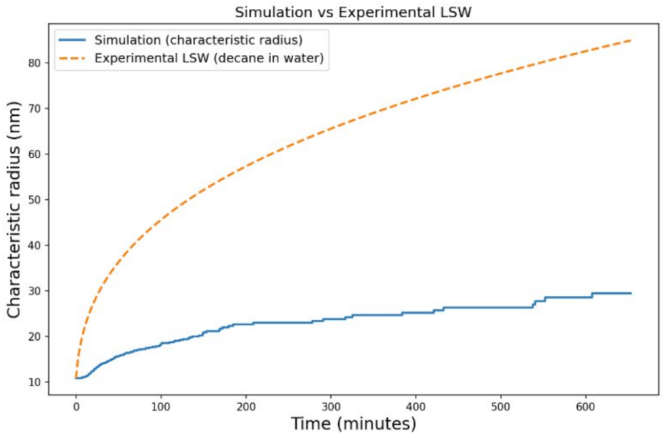


Figure 10: Comparison of the mean particle radius $\langle r \rangle$ evolution from our Ostwald ripening simulation (solid blue) with experimental LSW behavior reported for decane-in-water systems.[18] Simulation radii are converted using a scale of 1 simulation radius = 10 nm, and time is mapped using $\Delta t = 0.9761$ s per step.

Figure 10 shows the simulated growth rate is systematically slower than experiment, reflecting the simplified conversion of parameters from the LSW theory into a probabilistic approach. The enhanced mass transport and continuous solute redistribution accelerate coarsening relative to the discrete Monte Carlo dynamics. Although there is a large discrepancy in growth rates, our simulation follows the $\langle r \rangle^3$ proportional to t relationship. Thus, introducing size-dependent diffusion or transport-limited attachment kinetics could account for enhanced mass transfer in experimental systems. Of course, calibrating the molecular flux and diffusivity directly to experimentally measured values, rather than prescribing them as constants, would significantly reduce error.

3.4 Runtime and Code Scaling

Figure 11 shows the runtime after running single simulations for 5,000 time steps using different numbers of particles to show how the runtime scales with the number of particles in the simulation. For these simulations, weighting was set to `True`. The simulations were only carried out to 5,000 time

steps to allow for simulations to finish within a reasonable amount of time. For a given simulation, we find that the time per time step decreases as more and more particles are dissolved. Thus, by computing the first 5,000 time steps (when most of the particles are still alive) we can assess the efficiency of our method.

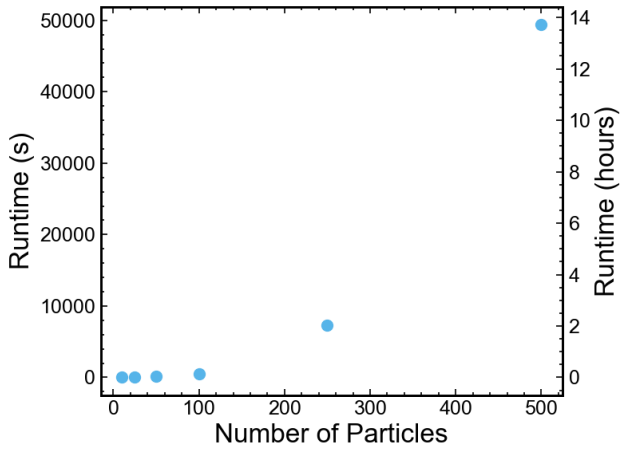


Figure 11: Run time as a function of number of particles run in the 5,000 time step simulation.

Figure 11 clearly shows an exponential increase in the run time as the number of particles is increased. Compared to other methods for modeling Ostwald ripening, our approach is substantially slower. For example, a study using phase-field modeling with a finite-difference framework simulated approximately 100,000 particles and reported runtimes of about three weeks on a personal computer with relatively modest specifications, over roughly 6,000–9,000 time steps [9]. By contrast, extrapolating our method to a 100,000-particle system suggests that a 5,000-timestep simulation would require on the order of one hundred years of computational time. Although the method exhibits poor scaling, it enabled simulations of a manageable number of particles within reasonable runtimes, which was sufficient to evaluate its general accuracy relative to theory. Future work could include modifying the algorithm to run faster for larger number of particle simulations.

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4 Expanded Scope

In this section, we address issues and questions raised during the Q&A session of the oral presentation.

4.1 Nucleating where $r < r_{crit}$

One major point of contention was our application of Ostwald ripening during particle nucleation where all of the particles were under r_{crit} (unstable). As shown in Figure 5 our simulation predicted power law growth similar to simulations where particles were above and below r_{crit}. This suggests that Ostwald ripening would occur despite the particles all being unstable. This seems extremely unlikely. After reading through publications [18],[19] it was found that the r_{crit} will shift over time as smaller particles dissolve and larger ones grow. If $r < r_{crit}$ the Gibbs–Thomson effect dominates (their chemical potential is higher than that of the surrounding solution). This leads to dissolving of the particle into the environment. The solute concentration increases, driving r_{crit} down. Eventually, some particles will have $r > r_{crit}$ leading to growth.

For future applications, modeling the concentration of the solute to modify r_{crit} is paramount for correctly modeling Ostwald ripening.

4.2 Variability in Results

Here, we present results from five independent simulations, each consisting of 500 particles evolved over 50,000 timesteps. For these simulations, weighting was disabled (`False`) to enable simulations over longer timescales with a larger number of particles. Figure 12 shows the average number of particles remaining at each timestep, along with the standard deviation across the five simulations. When viewed at full scale, the results exhibit relatively little variability; however, the zoomed-in inset reveals that the simulations have not fully converged after only five simulations. We would expect that if we continued to increase the number of simulations the average would converge more.

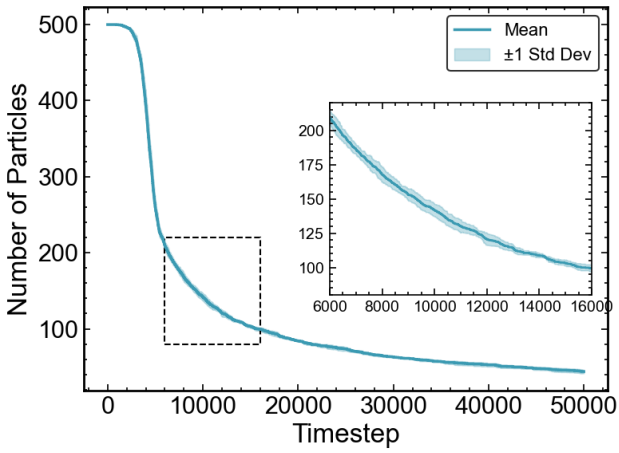


Figure 12: Results from running five independent 500-particle simulations for 50,000 timesteps. The number of particles remaining at each time step was average across the five simulations and the standard deviation was calculated. The inset plot shows a zoomed in view of the curve.

Figure 13 plots the mean, mean particle radius, $\langle r \rangle$, from the five simulations as a function of time step along with the standard deviation as shaded error bars and the power-law best-fit curve. Note, the initial drop seen in the first 5,000 time steps is due to the large number of particles starting under the critical radius that tend to shrink. Since there are so many more particles at the beginning of the simulation, the shaded error appears minuscule, but as there are fewer and fewer particles later in the simulation the standard deviation in mean radius becomes larger. Once again, running more Monte Carlo simulations would lead to a greater convergence in values.

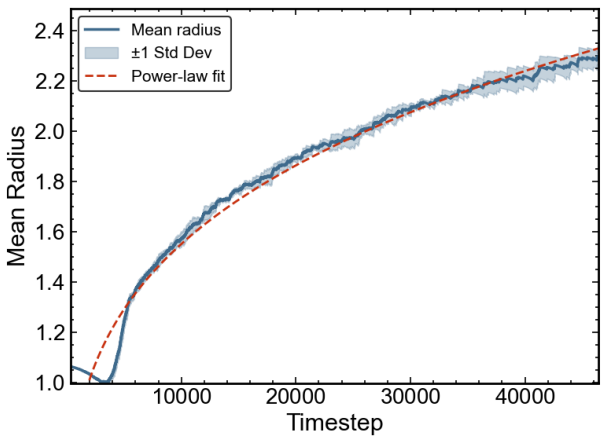


Figure 13: The mean, mean particle radius, $\langle r \rangle$, as a function of time step from the five 500-particle simulations. The standard deviation is given as shaded error bars, while the red dashed curve gives the best fit power-law curve for the mean data.

5 Conclusion & Future Work

In this work, we investigated a basic KMC simulation of Ostwald ripening and compared it to existing literature. In general, we found that the typical trends seen in systems undergoing Ostwald ripening such as the scaling of mean particle radius over time were conserved. Nonetheless, our results did not attempt to account for important processes like diffusion that could significantly impact results. To improve the physical relevance of our simulation, we can extend our model to include several additional factors.

Our model could benefit from incorporating diffusion kinetics into our simulation, allowing us to capture situations where diffusion occurs slowly relative to detachment and attachment. We could also choose the receiving particle based on a combination of the interface surface area and the particle's distance from its donor, as opposed to our current location-independent selection criteria. Additionally, the location of each particle could change over time due to particle interactions.

To simulate more complex interactions between particles and atoms, we can include atoms that have different volumes (e.g. half Fe and half Os), atoms with different full or partial charges (e.g polar vs nonpolar atoms), or atoms with different masses (using $F = ma$ to predict movement). To relate our simulation to specific real-world systems, we can run multiple KMC simulations (> 100) to estimate the values of specific parameters. We can also incorporate known parameters into our model to increase accuracy and generate time-evolution plots for metrics such as chemical potential or interfacial energy.

6 Contributions

- **Yari Armand:** Contributed to planning out the pseudo code. Co-wrote first iteration of Ostwald ripening code. Edited object-oriented Ostwald ripening code. Wrote code for comparing $\langle r_{sim} \rangle$ and $\langle r_{exprimental} \rangle$. Contributed to the writing of the presentation and reports (focused primarily on discussion).

- **Michael Flickinger:** Contributed to planning out the pseudo code. Ran large number of simulations used in the Results section and wrote code for data analysis and visualization of the outputted csvs. Primarily wrote introduction section on Ostwald ripening, and some areas of the Results & Discussion and Expanding Scope sections.

- **Vara Qi Gunananthan:** Contributed to problem definition, applying Monte Carlo to our scenario, simplifying assumptions, and planning out pseudo code. For final presentation and report: researched and wrote Monte Carlo introduction and assumptions sections, and helped with conclusions.

- **Logan Kuhn:** Team leader, contributed to planning out pseudo code. Wrote final iteration of Ostwald ripening code referenced in Approach Section and which was involved in plot creation. Contributed to the writing of the final presentation and report (primarily focused on Section 2: Approach). Ran multiple simulations of aforementioned code to provide data for report and presentation.

All authors agree to these contributions.

7 Collab HyperLink

Attached here is a link to access the Google Collab containing the entirety of code used in the scope of the project.

Google Collab Link

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